Process & Decision Documentation

Side Quests and A4 (Individual Work)

Keep this section brief, typically 2 to 4 sentences.

Focus on: adding a debug

Entry Header

Name: Suyao Liu

Role(s): Designer and coder

Primary responsibility for this work: coding and updating

Goal of Work Session

Briefly describe what you were trying to accomplish during this phase of the assignment.

Examples: adding a debug to my side quest 6

Tools, Resources, or Inputs Used

- Codex

GenAI Documentation

If GenAI was used (keep each response as brief as possible):

Date Used: Mar 23

Tool Disclosure: Codex

Purpose of Use: help with I gave it instruction on how to improve the game what to add, and it generated the code based on my command

Human Decision Point(s): Rejected some feature and double checking for error and play testing

Integrity & Verification Note: playtest

Scope of GenAI Use: It didn't give a playable update for level 2 at first

Limitations or Misfires: None

Summary of Process (Human + Tool)

Gave it instruction told it what we are doing and it gave me a fixed code

Decision Points & Trade-offs

I told it to modify how high the character can jump so it’s not too over the screen

I also found issue with level 2 when it first gave me, as the character can not jump

Verification & Judgement

I did multiple playtesting

Limitations, Dead Ends, or Open Questions

The character still jumps so high that it exceeds the screen but its not as high up as the first version

Appendix

create a debugg that allows you to toggle moon gravity on/off, and at least one other potential feature you might need in developing a game.

Worked for 1m 49s

Added an in-game debug system in sketch.js (line 220). It now tracks three toggles: moon gravity, invincibility, and probe/sensor visibility. The input handling and on-screen debug panel live in sketch.js (line 321) and sketch.js (line 615).

Controls:

- `: show/hide the debug panel
- G: toggle moon gravity on/off
- I: toggle invincibility on/off
- P: show/hide the player ground sensor and boar probes

I also wired the invincibility toggle into hazard/enemy damage checks in sketch.js (line 721) and sketch.js (line 740), and made sure gravity/debug visibility reapply when a level is built or changed in sketch.js (line 1161) and sketch.js (line 1248).

Verification: I reviewed the edited sections directly, but I could not run a node --check syntax pass because node is not installed in this workspace.

1 file changed

Undo

sketch.js

make sure there is a roof so it doesn't fly off the map too high up

Worked for 38s

Added an invisible roof collider so the player can’t fly above the map, including when moon gravity is enabled. The new ceiling is created in sketch.js (line 1324), registered in the globals at sketch.js (line 62), and both the player and boars now collide with it at sketch.js (line 1351) and sketch.js (line 1410).

I didn’t run an automated syntax check because node is not installed in this workspace.

1 file changed

Undo

sketch.js

make the roof higher just so the character doesn't go outside of the screen is enough

Worked for 17s

Raised the roof so it only stops the character from leaving the visible screen, instead of blocking them close to the map top. The ceiling height is now based on the viewport in sketch.js (line 172), and the ceiling sprite uses that value in sketch.js (line 1325).

I didn't run an automated syntax check because node is not installed here.

1 file changed

Undo

sketch.js

level 1 looks good but level 2 is unplayable the character can't jump the whole game is broken

Worked for 1m 51s

Fixed the level 2 breakage at the source in sketch.js (line 146). level2 had rows with lengths from 60 to 62 characters, so the map width, tile lookups, and collisions did not agree. I now normalize every level to a single width before the game uses it, which makes level 1 41 tiles wide and level 2 62 tiles wide consistently.

I also removed the fixed startup-based spawn assumption and made spawn/reset use the active level height in sketch.js (line 247), sketch.js (line 462), and sketch.js (line 1335). That makes the level setup less fragile going forward.

I couldn't run an automated playtest here, and node is not installed for a syntax check, but this addresses the concrete data inconsistency that would make level 2 unplayable.

1 file changed

Undo

sketch.js

update the read me file

I'm reading the current README.md first so I can update it to match the game as it exists now, including the new debug tools and the level/roof fixes.

Explored 2 files, 1 search

I've got the current README and the implemented features. I'm rewriting it so it reflects the actual controls, debug menu, level flow, and the recent stability fixes instead of leaving placeholder citation notes.

Created file

README.md +91-0

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## Project Title

GBDA 302 Sidequest 6: Kingsman


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## Student

- Name: `Suyao Liu`

- WatID: `s226liu`

- Student Number: `21069335`


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## Project Overview

Kingsman is a 2D side-scrolling platformer built with `p5.js`, `p5play`, and `p5.sound`. The player controls a king character who moves through two levels, rescues diamonds, avoids fire hazards, and defeats boar enemies.
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The project includes:

- Two playable levels with a level-select screen
- Parallax background layers
- Enemy patrol and hazard logic
- HUD for rescued diamonds and health
- Sound effects and looping background music
- A small in-game debug menu for development and testing

How To Run

1. Open `index.html` in a browser.
2. On the level-select screen, press `1` for Level 1 or `2` for Level 2.
3. Play using the controls listed below.

Controls

- `A` or `Left Arrow`: Move left
- `D` or `Right Arrow`: Move right
- `W` or `Up Arrow`: Jump
- `Space`: Attack
- `R`: Restart after winning or dying

Debug Controls

- `` ` ``: Toggle debug panel
- `G`: Toggle moon gravity on/off
- `I`: Toggle invincibility on/off
- `P`: Show or hide collision probes/sensors

Gameplay Features

- The player can run, jump, attack, take damage, and die.
- Boar enemies patrol platforms, turn around at edges, react to hazards, and can be defeated.

- Fire hazards damage the player and destroy boars.
- Diamonds increase the rescued count and are required to complete each level.
- A HUD shows rescued diamonds, current level, and remaining health.
- The game includes win, death, and restart states.

Technical Notes

- The game uses manual physics stepping for more stable pixel-art rendering.
- Level data is normalized so each tilemap row has a consistent width. This prevents collision and camera errors, especially in Level 2.
- An invisible ceiling prevents the player or enemies from leaving the top of the playable screen area.
- The debug menu was added to help test gravity, damage behavior, and collision probes during development.

Assets

- `assets/kinghuman_sprites.png`: player sprite sheet
- `assets/boar_sprites.png`: enemy sprite sheet
- `assets/diamond.png`: collectible sprite sheet
- `assets/fireSpriteSheet.png`: hazard animation sprite sheet
- `assets/bg-1.png`
- `assets/bg-2.png`
- `assets/bg-3.png`
- `assets/groundTile.png`
- `assets/groundTileDeep.png`
- `assets/platformLC.png`
- `assets/platformRC.png`
- `assets/wallL.png`
- `assets/wallR.png`
- `assets/bitmapFont.png`
- `assets/jump sound.mp3`
- `assets/attacking.mp3`
- `assets/hurt.mp3`

- `assets/coin.mp3`
- `assets/gameover.mp3`
- `assets/bgmusic2.mp3`

Libraries

- `p5.js`
- `p5play`
- `planck.js`
- `p5.sound`

References

- https://pixelfrog-assets.itch.io/kings-and-pigs
- https://pixabay.com/sound-effects/search/retro%20game%20music%20end/

The README rewrite is done. I'm doing a quick readback to make sure the formatting is clean and the debug controls are documented correctly.

Explored 1 file

Worked for 31s

Updated `README.md` to match the current project. It now documents the game overview, run instructions, player controls, debug controls, gameplay features, technical notes, assets, libraries, and references.

I also cleaned up the old placeholder citation section and corrected the title typo from "Sidequset" to "Sidequest."